Slices and String Methods ✂️

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Recall we can access parts of strings

We can access individual pieces of data inside a string using the `[]` (brackets)

Remember, we start counting from 0!

C	h	r	i	s
0	1	2	3	4

```
name = "Chris"

print(name[0], name[1], name[2], name[3], name[4])
```

```
C h r i s
```

Slices

We can access multiple pieces of data inside a string using the `[]` (brackets) like so:

`my_str[start:stop]` **where** `start` **is inclusive and** `stop` **is exclusive!**

- `start` **defaults to 0 if it is not provided!**
- `stop` **defaults to the length of the string if it is not provided!**

```python
name = "Christopher Martin"

print(name[5:9])
print(name[:5])
print(name[12:])
```

```
toph
Chris
Martin
```

More Advanced Slices

The indices we use when slicing can be positive or negative

- Indices that are positive are relative to the start of the String
- Indices that are negative are relative to the end of the String

C	h	r	i	s	t	o	p	h	e	r		M	a	r	t	i	n
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
-18	-17	-16	-15	-14	-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1

```python
name = "Christopher Martin"

print(name[-6:]) # Try to isolate everything but the first and last character!
```

More String behaviors

As we discussed, Strings have methods that let us inspect and modify their data!

Method	Description
`my_str.lower()`	Returns `my_str` with all alphabetic characters in lowercase
`my_str.upper()`	Returns `my_str` with all alphabetic characters in uppercase
`my_str.count(sub_str)`	Returns the number of times the specified `sub_str` occurs in `my_str`
`my_str.index(sub_str)`	Returns the `int` position as an index of the **first** occurance of the specified `sub_str` if it occurs anywhere in `my_str`, otherwise raises a `ValueError`

Even more String behaviors!

The `methods` **mentioned today and those discussed last week only represent a small selection of the possible String methods**

Strings have about `50` different methods they can execute by default!

Of these, there are many with different, more specific capabilities

We will see some of them later, but most won't be required for you to know

You can accomplish anything you'd need with slicing and the methods we have discussed in our slides!